AmirPouya MOEINI

Ph.D. Student, University of Cambridge, Trinity Hall

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Education

Ph.D. in Engineering (Information Engineering) Oct 2023 – Present

University of Cambridge, UK | Probabilistic Systems, Information, and Inference Group (PSI^2)
Supervisor: Prof. Albert Guillén i Fàbregas
Advisor: Prof. Ramji Venkataramanan

B.Sc. in Computer Engineering and Mathematics Sep 2018 – Jun 2023

Sharif University of Technology, Iran
Thesis: *Information-Theoretic Bounds for Online Learning and Sequential Prediction*
Supervisor: Prof. Mahdieh Soleymani Baghshah

Research Profile

Ph.D. research: Deriving achievability and converse bounds on capacity and the reliability function under mismatched and universal decoding, using large deviations and combinatorial methods.

Broader interests: High-dimensional probability, statistical learning theory, information-theoretic methods in inference, and probabilistic combinatorics.

Publications

Journal papers

[J1] The Expurgated Error Exponent is Not Universally Achievable arXiv
A. Moeini, M. Dalai, and A. Guillén i Fàbregas, submitted to *IEEE Transactions on Information Theory*.

Conference papers

[C7] Universal Decoding for Unequal Error Protection arXiv
A. Moeini and A. Guillén i Fàbregas, submitted to *IEEE Information Theory Workshop (ITW)*, 2027.

[C6] An Upper Bound to the Undetected Error Exponent with Erasure Decoding at Positive Rates arXiv
A. Moeini, M. Dalai, and A. Guillén i Fàbregas, submitted to *IEEE Information Theory Workshop (ITW)*, 2027.

[C5] A Counterexample to the Universal Achievability of the Expurgated Error Exponent
A. Moeini, M. Dalai, and A. Guillén i Fàbregas, submitted to *IEEE Information Theory Workshop (ITW)*, 2027.

[C4] Random Gilbert-Varshamov Codes for Joint Source-Channel Coding IEEE Xplore
A. Moeini and A. Guillén i Fàbregas, *IEEE International Symposium on Information Theory (ISIT)*, 2026.

[C3] Two-Class Joint Source-Channel Coding: Expurgated Exponents with i.i.d. Distributions DOI
A. Moeini and A. Guillén i Fàbregas, *International Zurich Seminar on Information and Communication (IZS)*, 2026.

[C2] Dual-Domain Exponent of Maximum Mutual Information Decoding IEEE Xplore
A. Moeini and A. Guillén i Fàbregas, *IEEE Information Theory Workshop (ITW)*, 2025.

[C1] Class-Based Expurgation Attains Csiszár's Expurgated Source-Channel Exponent IEEE Xplore
A. Moeini and A. Guillén i Fàbregas, *IEEE Information Theory Workshop (ITW)*, 2025.

Honours and Awards

2023 **Cambridge International Full Scholarship** Cambridge, UK
2022 **Summer Research Fellowship** *Aalto University* Espoo, Finland
2018 **Gold Medal** *International Olympiad on Astronomy and Astrophysics (ranked 9th overall)* Beijing, China
2017 **Gold Medal** *National Olympiad on Astronomy and Astrophysics* Iran

Academic Visits

Universitat Politècnica de Catalunya (UPC), Barcelona, Spain October 2026
Department of Signal Theory and Communications

University of Brescia, Brescia, Italy September 2025, September 2026
Department of Information Engineering, visiting Dr. Marco Dalai

National University of Singapore, Singapore October 2025
Institute of Data Science – School of Computing, visiting Dr. Jonathan Scarlett

Academic Service

Reviewer: *IEEE Transactions on Information Theory*; IEEE International Symposium on Information Theory (ISIT); International Zurich Seminar on Information and Communication (IZS).

Undergraduate Research

Probabilistic Machine Learning Group, Aalto University, Finland Jul 2022 – Oct 2022

- Approximate Bayesian Computation in the Presence of Outliers
- Supervisor: Prof. Samuel Kaski

- Developed computationally efficient methods to improve the robustness of Approximate Bayesian Computation under outliers and model misspecification.

Machine Learning Laboratory, Sharif University of Technology, Iran Oct 2021 – Mar 2023

- Information-Theoretic Bounds for Online Learning Problems
- Supervisors: Prof. Mahdieh Soleymani Baghshah and Dr. Hassan Hafez-Kolahi

- Introduced information-theoretic tools for online learning with adversarial data and derived performance upper bounds via conditional mutual information.

Teaching Experience

Supervision University of Cambridge

- Information Theory and Coding (3F7), Michaelmas 24–26
- Data Transmission (3F4), Lent 25, 26
- Mathematics (1P4), Easter 26, Michaelmas 26
- Mathematical Methods (2P7), Michaelmas 26 – Easter 27

Teaching Assistant Sharif University of Technology

- Information Theory and Coding*, Fall 22
- Theoretical Machine Learning*, Fall 22
- Statistical Machine Learning*, Spring 22
- Mathematical Statistics, Fall 21
- Data Structures and Algorithms, Spring 21
- Linear Algebra, Spring 21
- Regression Analysis, Spring 21
- Artificial Intelligence, Fall 21, Spring 22

*: Graduate-level (Ph.D.) Course

Selected Courses

University of Cambridge

Computational statistics and machine learning, Advanced information theory and coding

Sharif University of Technology

Probability

Information theory and coding*, Information-theoretic methods in high-dimensional statistics*, High-dimensional probability*, Graphical models*, Stochastic processes, Bayesian methods in statistics*, Index coding

Mathematics

Real analysis, Algebra, Linear algebra, Applied linear algebra, Probability & statistics

Machine Learning

Machine learning theory*, Artificial intelligence, Computer simulation, Modern information retrieval

Computer Science

Design algorithms, Data structures and algorithms, Compiler design, Database design

Audited Courses

Probabilistic combinatorics, Entropy methods in combinatorics, Ramsey theory, Random discrete structures, The Ising model, Percolation, Stochastic calculus and applications to finance (*Cambridge*)

Skills

Programming: Python, C++, SQL, Julia, R, Java

Scientific computing and ML: PyTorch, NumPy, pandas, SciPy, scikit-learn

Tools: Git, Linux, LaTeX

Work Experience

Data Scientist Intern | *Comma AI-Med* Mar 2023 – Jun 2023

Developed a GAN-based pipeline that synthesises tumours in mammography images to augment tumour-detection data.

Data Engineer Intern | *Institute for Research in Fundamental Sciences (IPM)* Jun 2020 – Sep 2020

Built a data pipeline and a cloud-prediction model for weather-station measurements from Iran's National Observatory.